DO NOW — RECALL

Hash tables and collisions

Name: _____ Class: _____ Date: _____

1. In an undirected graph, why does edge A–B appear in the adjacency list of both A and B?

2. In a BST, 18 is smaller than root 25. Which direction do you move?

3. What are valid indexes of a five-element C# array?

4. Calculate 17 MOD 6.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Because traversal is permitted both ways.

2. Model answer

Left.

3. Model answer

0, 1, 2, 3, 4.

4. Model answer

5, since 17 = 2×6 + 5.

Vocabulary — know these terms

hash function — Maps a key to an initial bucket index.

collision — Two keys map to the same initial bucket.

linear probing — Checks successive buckets, wrapping around when needed.

load factor — The proportion of occupied table buckets.

clustering — A sequence of occupied buckets increasing probe lengths.

NEW LEARNING

Read, annotate and explain

Calculate the index

9 MOD 7 = 2, so put 9 into bucket 2.

Detect a collision

16 MOD 7 is also 2. The bucket is occupied, so move to the next bucket, 3.

Probe until empty

23 also hashes to 2; buckets 2 and 3 are full, so place it in 4.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Insert 6, 11, 16 into five buckets using h(k)=k MOD 5 and linear probing.

Predict and explain

Insert 6 at index 1, because 6 MOD 5 = 1.

Check and extend

11 collides at 1, so store it at 2.

Check and extend

16 collides at 1 and 2, so store it at 3.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Using h(k)=k MOD 7, find the starting index for 24.

Question 2

Insert 4, 9, 14 into five empty buckets using h(k)=k MOD 5 and linear probing. Give final indexes.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain the effect of a high load factor on linear probing.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A hash collision means two different keys have the same starting bucket.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.